

C Revision Map

This document condenses core C programming concepts into a quick revision structure.

1. Basics

Data types, variables, operators. Understand memory size, type modifiers, and operator precedence.

2. Control Flow

if-else, switch statements. Focus on condition evaluation and fall-through behavior.

3. Loops

for, while, do-while. Practice iteration patterns and loop control (break, continue).

4. Functions

Function declaration, definition, recursion, parameter passing (call by value).

5. Arrays & Strings

1D/2D arrays, string manipulation without library functions, traversal techniques.

6. Pointers

Pointer basics, dereferencing, pointer arithmetic, arrays vs pointers, double pointers.

7. Structures & Unions

Struct creation, nested structs, memory differences between struct and union.

8. File Handling

File operations (fopen, fclose, fprintf, fscanf, fread, fwrite).

9. Dynamic Memory

malloc, calloc, realloc, free. Memory lifecycle and leaks.